

TG-088

Motor Trip and Electrical Fault Investigation

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Applicable unit	Unit 5
Asset class	motor
Classification	Internal

1 Purpose

This guide supports investigation of motor trips on medium-voltage drives, including identification of the assets that must be inspected before a restart is attempted.

2 Trip Classification

Protection element	Typical cause	Restart permitted before inspection
Thermal overload	Sustained overload or cooling loss	No
Instantaneous overcurrent	Winding or cable fault	No
Earth fault	Insulation breakdown	No
Undervoltage	Supply disturbance	Yes, after supply confirmed stable
Vibration trip	Mechanical fault in driven equipment	No

Table 2-1 — Trip cause by protection element

WARNING Where the protection element is not positively identified, treat the trip as a winding or cable fault and do not attempt a restart.

3 Related Assets to Inspect

A motor trip is rarely isolated to the motor. The following assets shall be inspected before a restart is authorised, in the order listed:

1. The driven equipment, for a mechanical cause that produced the electrical symptom.

2. The coupling and its guard, for misalignment, wear or contact damage.
3. The motor bearings, both drive end and non-drive end.
4. The cooling path, including any air filter, fan and cooling water circuit.
5. The supply cable and terminations, for insulation damage or moisture ingress.
6. The upstream switchgear and protection relay event log.
7. Any upstream process asset whose alarm preceded the trip within 30 minutes.

NOTE Where the driven equipment is a pump, inspect the suction side before concluding a motor fault. A suction-side restriction can present as a motor overload trip. Refer to TG-051.

4 Restart Authorisation

Restart is authorised by the responsible electrical engineer following completion of §3 and confirmation of insulation resistance above 100 megohms at 500 V DC. Isolation under SI-009 applies to all inspection work in §3.